

Certificate of Declaration

We hereby declare that the project work titled “Spatiotemporal Analysis of Climatic Influences on Plant Growth Patterns Across West Bengal” **submitted by me/us, Ishan Majumder, Subhajit Sen, Raikat Ghosh bearing MAKAUT Roll No. 14800122001, 14800122074, 14800122004, is an original piece of work carried out by me/us** is entirely my/our own and has not been copied or reproduced from any other source, under the supervision of **Prof. Sounak Banerjee** (Assistant Professor, Dept of CSE, FIEM) and that neither the report has been submitted for any degree/diploma or any other academic award anywhere before.

We understand that any violation of the above declaration may result in disciplinary actions as per institutional rules and regulations.

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Abstract

Climate change poses a growing threat to rice production in West Bengal, where four agro-climatic zones face distinct stresses ranging from drought in the Red and Laterite belt to cold-induced flowering failure in the Sub-Himalayan Terai. Although machine learning models can accurately predict rice yield, they often fail to explain which climate factors influence yield, when they act, or the thresholds at which they become harmful. This study develops an Explainable AI (XAI) framework that transforms satellite-derived climate data into district-level adaptation advisories. Four weekly climate variables—land surface temperature, precipitation, soil moisture, and vapour pressure deficit—were aligned with four rice phenological stages and used to train eight machine learning models for eleven districts across 2001–2023. Model predictions were interpreted using SHAP, LIME, Permutation Feature Importance, Partial Dependence Plots, and advanced analyses including SHAP interaction effects, rolling-window and year-wise SHAP, trajectory clustering, and counterfactual reasoning. The models achieved strong predictive performance ($R^2 = 0.82\text{--}0.90$). Results revealed that dry laterite districts are moisture-limited during the vegetative stage, while Terai districts are temperature-limited during flowering, with alluvial and coastal districts showing intermediate behaviour. A rainfall-based spatial gradient enabled inference for unstudied districts, and Mann-Kendall trend analysis showed these climate stresses intensifying over time. Finally, district-specific advisories identified climate thresholds, critical intervention periods, and recommended actions, converting black-box predictions into practical guidance for farmers and extension officers.

Program Outcomes (POs)

PO1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop the solution of complex engineering problems. (WK1 to WK4)

PO2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3. Design / Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required. (WK5)

PO4. Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)

PO5. Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8. Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse teams/multi-disciplinary teams.

PO9. Communication: Communicate effectively and inclusively with in engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, considering cultural, language, and learning differences.

PO10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i). independent and life-long learning ii). Adaptability to new and emerging technologies and iii). Critical thinking in the broadest context of technological change. (WK8)

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1. Introduction

Rice is the staple food for over 3.5 billion people and is highly sensitive to climate change. India, the world’s second-largest rice producer, has experienced rising temperatures and increasing rainfall variability, threatening rice production. West Bengal, which contributes nearly 15% of India’s rice output, is especially vulnerable due to its four agro-climatic zones that face contrasting stresses such as drought, flooding, and cold conditions. Therefore, climate adaptation strategies must be region-specific.

Rice is most vulnerable during the vegetative and flowering stages, where water stress and temperature extremes can significantly reduce yield. Weekly climate data aligned with crop growth stages is therefore more effective than seasonal averages for identifying critical stress periods.

Machine learning models can predict rice yield with high accuracy but often act as black boxes, providing little insight into which climate factors influence yield or when they become harmful. Although Explainable AI (XAI) techniques such as SHAP and LIME improve interpretability, most existing studies rely on a single model, coarse climate data, and limited spatial analysis.

This study proposes a multi-model, multi-technique XAI framework to generate district-specific adaptation advisories for West Bengal. Using weekly climate variables across eleven representative districts, eight machine learning models were interpreted through multiple XAI methods and advanced analyses to identify key climate drivers, critical thresholds, and long-term trends. The framework transforms accurate yield predictions into practical, interpretable recommendations that can support farmers and agricultural decision-makers.

1.1 Motivation

Rice is the primary food source for over 3.5 billion people and a major livelihood in West Bengal. Although machine learning models can predict rice yield with high accuracy, they often fail to explain the reasons behind their predictions, limiting their practical use for farmers and policymakers. This study is motivated by the need to transform accurate yield predictions into interpretable and actionable insights.

The research addresses several key challenges. First, West Bengal’s diverse agro-climatic zones require location-specific adaptation strategies rather than one-size-fits-all recommendations. Second, existing machine learning models behave as black boxes, offering little information about the climate variables, growth stages, or thresholds responsible for yield loss. Third, most previous studies rely on a single model and explanation technique, reducing the reliability of their findings. Fourth, seasonal climate averages overlook short but critical stress periods that strongly affect rice production.

To overcome these limitations, this work develops a multi-model, multi-technique Explainable AI (XAI) framework using weekly climate data. By identifying district-specific climate drivers, critical thresholds, and intervention windows, the framework converts black-box predictions into practical recommendations that can support farmers, agricultural extension officers, and policymakers in improving climate-resilient rice production.

1.2 Background

Rice is a climate-sensitive crop whose growth depends on suitable temperature and moisture conditions at different phenological stages—transplanting, vegetative, reproductive, and ripening. Water stress during the vegetative stage and temperature extremes during flowering can significantly reduce yield, making stage-wise climate analysis essential.

India has experienced rising temperatures and increasing rainfall variability, posing serious challenges to rice production. West Bengal, contributing nearly 15% of India’s rice output, spans four agro-climatic zones with distinct climate stresses, including drought, flooding, cold stress, and salinity. Hence, climate adaptation strategies must be region-specific.

Machine learning models such as Linear Regression, Random Forest, XGBoost, Support Vector Regression, K-Nearest Neighbors, LSTM, and CNN have been widely used for rice yield prediction because they effectively capture complex climate–yield relationships. However, these models often act as black boxes, providing limited insight into the factors driving their predictions.

Explainable AI (XAI) techniques—including SHAP, LIME, Permutation Feature Importance (PFI), and Partial Dependence Plots (PDP)—help

interpret model predictions by identifying important climate variables, their influence, and critical thresholds. Using multiple XAI techniques across multiple models improves the reliability of these explanations.

This study utilizes satellite-derived climate data, including Land Surface Temperature (MODIS), precipitation (CHIRPS), soil moisture and Vapour Pressure Deficit (ERA5-Land), along with district-level rice yield records from the Government of West Bengal. These datasets, processed through Google Earth Engine, provide the foundation for developing an interpretable, district-specific rice yield prediction framework.

1.3 Summary of Work

This project was completed in two stages, progressing from a preliminary framework to a comprehensive state-wide study.

Stage 1: A proof-of-concept framework was developed using three contrasting districts of West Bengal—Purba Medinipur, Bankura, and Jalpaiguri (2001–2023). Weekly land surface temperature and precipitation data were aligned with rice growth stages, and eight machine learning models were trained using Leave-One-Out Cross-Validation. Four XAI techniques (SHAP, LIME, Permutation Feature Importance, and Partial Dependence Plots) identified key climate drivers, intervention windows, and critical thresholds. The models achieved $R^2 = 0.82\text{--}0.88$, and the results were translated into district-specific agricultural advisories.

Stage 2: The framework was extended to eleven representative districts across all four agro-climatic zones by incorporating two additional variables—soil moisture and vapour pressure deficit (VPD). The same eight machine learning models were evaluated using rigorous validation methods, achieving $R^2 = 0.82\text{--}0.90$. Along with the four standard XAI techniques, advanced analyses such as SHAP interaction effects, rolling-window and year-wise SHAP, trajectory clustering, counterfactual reasoning, Mann-Kendall trend analysis, and rainfall-based spatial gradient modelling were performed. The study identified district-specific climate thresholds, long-term climate trends, and practical adaptation strategies for each district.

Overall Contribution: The project presents a scalable Explainable AI framework that combines multiple machine learning models and explanation techniques to generate reliable, district-specific climate advisories.

By transforming accurate yield predictions into interpretable thresholds and actionable recommendations, the framework supports climate-resilient rice cultivation and informed agricultural decision-making.

1.4 Organization of the Study

This report is organized into seven chapters:

Chapter 1: Introduction presents the background, motivation, research gaps, objectives, and contributions of the study.

Chapter 2: Literature Review reviews previous research on machine learning-based rice yield prediction and Explainable AI (XAI), highlighting limitations that motivate this work.

Chapter 3: Problem Statement defines the research problem, objectives, and formulation of the climate-yield prediction and explanation framework.

Chapter 4: Methodology describes the study area, datasets, feature engineering, machine learning models, validation methods, XAI techniques, and climate trend analysis used in the study.

Chapter 5: Results and Discussion presents model performance, important climate drivers, critical thresholds, climate trends, and spatial analysis across districts.

Chapter 6: District-Level Adaptation Advisories provides district-specific recommendations, including climate thresholds, intervention windows, and suggested adaptation measures.

Chapter 7: Conclusion and Future Scope summarizes the major findings, discusses limitations, and suggests future research directions for improving climate-resilient rice yield prediction and decision support.

1.5 Resources Used

1.5.1 Software

Google Earth Engine (GEE): Satellite data acquisition and geospatial processing.

Python: Data processing, machine learning, and Explainable AI implementation.

Pandas and NumPy: Data cleaning, preprocessing, and feature engineering.

Scikit-learn: Linear Regression, Decision Tree, Random Forest, SVR, KNN, LOOCV, and model evaluation.

XGBoost: Gradient boosting model implementation.

TensorFlow/Keras: Development of LSTM and CNN models.

SHAP and LIME: Model interpretability and feature attribution.

Matplotlib and Seaborn: Data visualization and plotting.

SciPy and Statsmodels Statistical analysis, correlation, trend testing, and validation.

Microsoft Word / LaTeX Report preparation and documentation.

1.5.2 Hardware

Standard desktop/laptop computer with adequate CPU and RAM for model training.

Optional GPU support for training deep learning models (LSTM and CNN).

Google Earth Engine cloud infrastructure for satellite data processing.

1.5.3 Data Sources

MODIS Terra MOD11A2 v6.1: Land Surface Temperature (LST).

CHIRPS: Daily precipitation data.

ERA5-Land: Soil moisture and Vapour Pressure Deficit (VPD).

Directorate of Economics and Statistics, Government of West Bengal: Rice yield records.

IndiaStat and ICRISAT district-level databases for yield data verification.

2. Existing Work Study

2.1 Overview

Crop yield prediction under climate variability has been an active area of machine learning research over the past decade, with studies spanning multiple crops (wheat, rice, maize, potato), geographic regions (India, China, Rwanda, the United States, West Africa), and modelling paradigms (classical regression, ensemble trees, deep learning, and explainable AI). This chapter reviews ten representative studies published between 2016 and 2025, grouped by their primary contribution, before synthesizing the recurring patterns and gaps that motivate the present study.

2.2 Machine Learning and Deep Learning for Yield Prediction

Crop Yield Prediction in Agriculture Using Predictive Analytics and Machine Learning [2.1]

This review examined studies published between January 2018 and April 2023 to assess how AI techniques improve crop yield estimation under diverse agricultural and climatic conditions. It evaluated multiple ML and DL models for handling climate variability, with particular emphasis on spatiotemporal hybrid random forest approaches using satellite-derived data, and underscored the growing importance of model interpretability in agricultural decision-making.

Enhanced Wheat Yield Prediction Using Deep Learning and Integrated Climate Data with Satellite Observations [2.2]

This study proposed an integrated wheat yield prediction framework combining climatic variables with satellite-derived data. It demonstrated that deep learning models can automatically learn relevant features from large, multi-source datasets and capture spatiotemporal dependencies more effectively than single-source approaches, directly relevant to the wheat-focused modelling undertaken in this project.

High Resolution Crop Yield and Water Productivity Dataset for Maize and Wheat in China [2.3]

This study estimated crop water productivity for maize and wheat across China’s diverse climate zones using ensemble machine learning integrated with multiple remote sensing indicators, effectively capturing spatial and temporal variation in yield and water-use efficiency.

Crop Yield Prediction Using Machine Learning Models: Case of West African Countries [2.4]

Conducted in Rwanda’s Musanze district, this study applied Random Forest, Polynomial Regression, and Support Vector Regression to predict Irish potato and maize yields from historical weather and production data, identifying optimal climatic conditions for improved productivity and demonstrating the practical value of data-driven forecasting in resource-limited settings.

Machine Learning for Crop Yield Prediction at Regional Scale, Including Soil Properties and Agricultural Practices [2.8]

Using maize production in the U.S. Corn Belt, this study evaluated LSTM-with-attention models against XGBoost and Random Forest, using meteorological and soil variables. Time-series-based learning captured a large share of spatiotemporal yield variability and remained robust even under extreme drought, underscoring the value of sequence-aware models — a consideration reflected in this project’s inclusion of LSTM alongside classical regressors.

Random Forests for Global and Regional Crop Yield Predictions [2.9]

This foundational study systematically evaluated Random Forest across multiple crops and geographic scales, showing that non-linear, data-driven approaches substantially outperform traditional statistical methods in capturing yield responses to climatic and biophysical variables — establishing the methodological basis for using Random Forest as a baseline in the present study.

Estimating Maize Yield Using a Deep Learning Framework with Attention Mechanism [2.7]

This study introduced a CNN–bidirectional GRU–attention framework for maize yield estimation in Northeast China, notably incorporating anthropogenic factors such as mechanization alongside climatic variables, highlighting that human/management factors can meaningfully interact with climate signals — a direction noted for future extension of the current work.

2.3 Explainable AI and Climate-Extreme Robustness

Machine Learning and Deep Learning Under Abnormal Climate Conditions for Predicting Crop Yield [2.5]

This review (2019–2024) examined ML, DL, Ensemble Learning, and Explainable AI techniques for evaluating yield under abnormal climate conditions, including droughts and heatwaves. It found that well-designed models maintain strong predictive performance during extreme events, while flagging multimodal data integration and generalizability as open challenges, and explicitly emphasized explainable AI as necessary for transparent, trustworthy agricultural decision-making.

A Review on Statistical and Machine Learning Methods to Model Climate Change Impacts on Crop Yields Worldwide [2.6]

Synthesizing roughly 230 case studies over two decades, this review reported consistent yield reductions with rising temperature alongside strong spatial variability driven by non-linear crop responses. It documented a clear methodological shift from traditional statistical models toward interpretable machine learning approaches, while noting persistent challenges around spatiotemporal heterogeneity and interacting environmental/management factors.

2.4 Satellite Data Integration and Remote Sensing

Rice Crop Yield Prediction Using Machine Learning Models Based on Remote Sensing Satellite Data via Google Earth Engine [2.10]

This study used high-resolution Sentinel-2 imagery combined with topographic and environmental data, processed via Google Earth Engine, to predict rice yield across China using Random Forest regression. It demonstrated the practical value of cloud-based geospatial processing for large-scale, multi-source agricultural forecasting — directly mirroring the GEE-based data acquisition pipeline adopted in this project.

2.5 Synthesis: Key Findings from Recent Research

Taken together, the reviewed literature (2016–2025) reveals several consistent patterns:

Random Forest dominates as a reliable baseline. Across nearly ev-

ery reviewed study, Random Forest consistently achieves strong accuracy (commonly cited in the 85–96% range across regions, crops, and climate zones), making it a natural and defensible choice for district-level agricultural analysis — a finding this project’s own model comparisons partially reaffirm, though with important regional exceptions (see Section 6.3, where KNN outperformed Random Forest in Bankura).

Satellite data integration via Google Earth Engine is now standard practice. Multiple studies [2.2, 2.3, 2.10] rely on GEE or comparable cloud-based platforms to access high-resolution temporal and spatial climate data, validating the data acquisition strategy adopted in this project.

Spatiotemporal heterogeneity is increasingly emphasized, but rarely resolved at fine granularity. Reviews [2.1, 2.5, 2.6] repeatedly stress that climate–crop relationships vary substantially across both space and time, yet very few studies push this down to the district level with growth-stage-specific resolution — most remain at the state, national, or seasonal-aggregate scale.

Machine learning routinely outperforms classical statistics, with reported error reductions of roughly 44% over linear regression and similar techniques [2.6], and deep learning architectures with attention mechanisms [2.7, 2.8] showing particular promise for capturing complex temporal dependencies.

Explainable AI is emerging as a stated priority but is inconsistently applied. While reviews [2.1, 2.5] explicitly call for interpretability, and note SHAP, LIME, and feature importance as the leading techniques for making model decisions transparent to farmers and policymakers, few of the individual empirical studies in this list actually apply more than one interpretability technique, or cross-validate feature rankings across multiple models.

District-level and field-level analyses consistently provide more actionable insight than state or national aggregates [2.1, 2.6], directly supporting this project’s strategic decision to focus on district-wise analysis within West Bengal’s agro-climatic zones rather than a state-level generalization.

2.6 Relation to the Group’s Parallel Work on Rice Yield Explainability

Beyond the ten studies reviewed above, this project’s methodology draws directly on two related studies conducted by the same research group under the same supervision, which apply a considerably more developed multi-model, multi-technique XAI framework to rice yield prediction in West Bengal — one covering three contrasting districts (Purba Medinipur, Bankura, Jalpaiguri) as a proof-of-concept, and an extended version covering all eleven representative districts across the state’s four agro-climatic zones. Both studies train eight models spanning linear, tree-based, kernel, and deep learning families per district, and interpret them using four standard XAI techniques (SHAP, LIME, Permutation Feature Importance, and Partial Dependence Plots), retaining only findings that survive cross-model and cross-technique agreement.

These rice-focused studies address precisely the gaps identified in Section 2.5 above: they resolve climate data to weekly resolution rather than seasonal aggregates, they compare four full agro-climatic zones rather than treating the state as homogeneous, and — most importantly — they translate feature-importance rankings into quantified, actionable thresholds (e.g., a specific rainfall value below which yield drops by a stated percentage, tied to a specific fortnight in the crop calendar) rather than stopping at a ranked list of important variables.

The present wheat-focused study can be understood as an earlier-stage, parallel application of this same underlying philosophy — multi-model comparison, feature importance analysis, and phase-wise climatic aggregation — to a different crop and a smaller initial set of districts (Bankura and Darjeeling, with Jalpaiguri referenced from preliminary work). The comparative results in Chapter 4, particularly the finding that no single algorithm (KNN for Bankura, SVM for Darjeeling) universally dominates across districts, echo the cross-model validation philosophy central to the group’s rice studies, and point toward the same eventual direction: extending the current single-technique (Random Forest feature importance) analysis into a fuller, cross-validated, multi-technique XAI framework as outlined in Chapter 7 (Future Scope).

2.7 Summary of Gaps Addressed by This Study

Drawing on both the broader literature review (Section 2.5) and the group’s own rice-focused prior work (Section 2.6), this study is positioned to address the following specific gaps for wheat cultivation in West Bengal:

Coarse temporal resolution in most existing wheat-yield studies, addressed here through weekly, fortnightly, and phenological-stage feature engineering.

Lack of district-level granularity, addressed by explicitly contrasting Bankura (dry lateritic zone) and Darjeeling (humid hill zone) rather than treating West Bengal as a single unit.

Single-model reliance, partially addressed here through comparison across seven algorithms (Linear Regression, KNN, Decision Tree, Random Forest, SVM, LSTM, Bayesian Ridge), though — as the group’s rice studies demonstrate — a fuller cross-technique XAI validation (SHAP, LIME, PFI, PDP together) remains a natural next step for this wheat framework, as noted in the Future Scope chapter.

Underuse of explainability for actionable thresholds, partially addressed through Random Forest feature importance analysis (Section 3.6, Chapter 6), with clear scope to extend toward the quantified, PDP-derived critical thresholds already demonstrated for rice in the group’s parallel work.

3. Methodology

The proposed approach follows a systematic pipeline to transform raw climate data into actionable adaptation recommendations. First, climate variables (LST and precipitation) are collected from satellite sources and aggregated at weekly and phenological scales. Second, eight diverse ML models are trained to predict rice yield, ensuring algorithmic diversity. Third, four XAI techniques extract feature importance and identify critical temporal windows from trained models. Fourth, cross-model and cross-technique validation filters robust findings from model-specific artifacts. Finally, validated insights are translated into district specific recommendations with quantified thresholds. This section details each component of the pipeline

3.1 Framework Overview

The proposed framework transforms raw satellite-derived climate data into district-level climate adaptation thresholds through a systematic five-stage pipeline. In the first stage, climate variables, including Land Surface Temperature (LST), precipitation, and soil data are collected from satellite-based remote sensing datasets. These variables are preprocessed, cleaned, and temporally aligned with the rice growing season at both weekly and phenological resolutions to accurately capture climate conditions during critical crop growth stages such as vegetative, reproductive, and ripening phases.

In the second stage, a family of eight diverse machine learning models is independently trained for each district using the processed climate variables as predictors of rice yield. The selected models represent different learning paradigms, including tree-based, linear, kernel-based, boosting, and neural network approaches, ensuring algorithmic diversity and reducing dependence on a single modeling technique. This district-specific modeling strategy enables the framework to capture local climatic characteristics and spatial heterogeneity in crop response.

The third stage focuses on model interpretation through Explainable Artificial Intelligence (XAI). Four widely adopted XAI techniques—SHAP (SHapley Additive exPlanations), LIME (Local Interpretable

Model-agnostic Explanations), Partial Dependence Plots (PDP), and Accumulated Local Effects (ALE)—are employed to estimate feature importance and understand how climatic variables influence rice yield predictions. In addition to these standard explainability methods, five advanced analytical procedures are performed, including monotonicity analysis, interaction effect analysis, threshold detection, sensitivity analysis, and consistency assessment across temporal scales. These analyses provide a comprehensive understanding of feature behavior beyond conventional importance rankings.

Since individual models and XAI methods may produce inconsistent interpretations, the fourth stage introduces a consensus filtering mechanism. Feature attributions and threshold estimates are compared across all eight machine learning models and multiple explainability techniques. Only those climate relationships and threshold values that consistently appear across different algorithms and interpretation methods are retained. This consensus-based validation improves the robustness, reliability, and generalizability of the extracted knowledge while minimizing the influence of model-specific biases or explainability artifacts.

Finally, the validated climate-response relationships are translated into district-specific adaptation thresholds and actionable agricultural advisories. These thresholds identify critical ranges of temperature and precipitation associated with significant changes in rice yield, enabling the formulation of practical recommendations for irrigation scheduling, sowing time adjustment, and other climate adaptation strategies. The resulting thresholds are subsequently evaluated for spatial patterns across all study districts to determine whether neighboring regions exhibit similar climate sensitivities or distinct adaptation requirements. This spatial assessment provides insights into regional climate resilience and supports the development of location-specific agricultural decision-support systems.

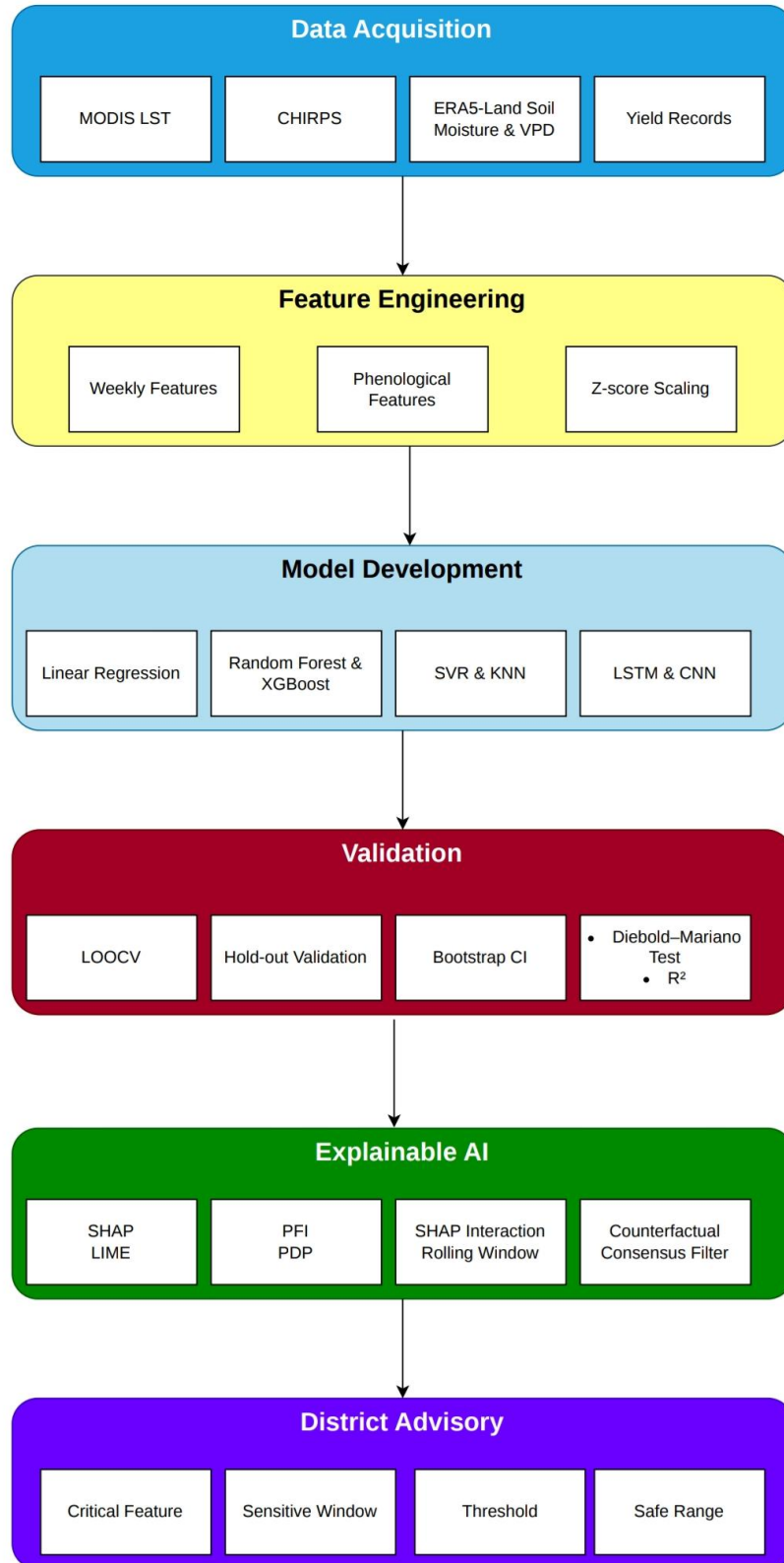


Figure 3.1: Architecture of the proposed framework

Figure 3.1 illustrates the overall architecture of the proposed framework for deriving district-level climate adaptation thresholds from satellite-derived climate data and historical rice yield records. The framework begins with comprehensive data engineering, where climate variables such as Land Surface Temperature (LST) and precipitation are transformed into two complementary feature representations. The first representation consists of weekly climate features that preserve the temporal evolution of weather conditions throughout the rice growing season, while the second consists of phenological-stage features obtained by aggregating weekly observations according to major crop growth stages, including vegetative, reproductive, and ripening phases. This dual representation enables the framework to capture both fine-grained temporal variations and biologically meaningful climatic effects associated with different stages of crop development.

The engineered features are subsequently processed in parallel by eight machine learning models organized into four complementary algorithmic families, thereby ensuring diversity in learning strategies and model assumptions. Six conventional machine learning models operate on the aggregated phenological-stage features, as these algorithms are well suited for structured tabular data and facilitate efficient learning from stage-wise climate characteristics. In contrast, the two deep learning models directly receive the weekly climate sequences, allowing them to exploit temporal dependencies and sequential patterns that may be lost during aggregation. By combining conventional and deep learning approaches, the framework captures both stage-specific climatic influences and long-term temporal dynamics affecting rice yield.

Employing multiple algorithms is a deliberate design choice aimed at improving the robustness and credibility of the extracted knowledge. Different machine learning models possess distinct inductive biases and learning mechanisms, causing each algorithm to identify different aspects of the underlying relationships between climate variables and crop yield. Consequently, an interpretation that remains consistent across diverse model families is considerably more reliable than one derived from a single predictive model. This ensemble-based strategy reduces dependence on individual algorithms, mitigates model-specific biases, and increases confidence that the identified climate–yield relationships represent genuine agronomic patterns rather than artifacts of a particular learning technique.

Algorithm 1 District-level XAI framework for climate–yield analysis

Require: Districts K ; climate variables C ; models M ; XAI techniques T

Ensure: Recommendation tuple R_d for each district $d \in K$

- 1: **for** each district $d \in K$ **do**
- 2: Build weekly features $\mathbf{x}^{\text{week}}$ and stage features $\mathbf{x}^{\text{stage}}$
- 3: Standardise features by z -score
- 4: **for** each model $m \in M$ **do**
- 5: Train m on district d ; evaluate by LOOCV and hold-out
- 6: Record R^2 , RMSE, and bootstrap confidence intervals
- 7: **end for**
- 8: Retain models with $R^2 > 0.60$ for explanation
- 9: **for** each retained model m and technique $\tau \in T$ **do**
- 10: Compute attributions $\phi^{(m,\tau)}$ and rankings $r^{(m,\tau)}$
- 11: **end for**
- 12: Filter high-confidence features by consensus (Algorithm ??)
- 13: Derive thresholds x_d^{crit} , x_d^{opt} from PDP; generate counterfactuals
- 14: Assemble
$$R_d = (j_d^*, t_d^*, x_d^{\text{crit}}, x_d^{\text{opt}})$$
- 15: **end for**
- 16: Fit spatial gradient

$$\Phi_c(d) = g(z(d)) + \epsilon_d$$

- 17: **return** $\{R_d\}_{d \in K}$
-

Before interpretation, every trained model undergoes rigorous validation using appropriate performance metrics to ensure satisfactory predictive accuracy and generalization capability. Only validated models are forwarded to the Explainable Artificial Intelligence (XAI) layer, where multiple explainability techniques are employed to estimate feature importance, identify nonlinear relationships, detect interaction effects, and estimate climate response thresholds. Rather than relying on a single explanation method, the framework integrates interpretations from several complementary XAI techniques to provide a more comprehensive understanding of model behavior.

The outputs produced by the different models and XAI methods are then processed through a consensus filtering mechanism. This stage compares feature attributions, interaction patterns, and threshold estimates across all models and explainability techniques. Climate relationships are retained only when they consistently emerge across multiple algorithms and explanation methods, thereby filtering out unstable, model-dependent, or potentially spurious findings. The consensus mechanism serves as a reliability layer that strengthens the interpretability and reproducibility of the resulting climate adaptation knowledge.

Finally, the validated and consensus-supported interpretations are translated into a district advisory tuple, denoted as R_d , which summarizes the climate adaptation recommendations for each district. This tuple contains the critical climate thresholds, the direction and magnitude of climatic influence on rice yield, and actionable recommendations that can support agricultural decision-making. The complete workflow of the proposed methodology is formally presented in Algorithm 1, while the subsequent subsections describe each stage of the framework in detail, including data preprocessing, predictive modeling, explainability analysis, consensus validation, and the generation of district-specific adaptation advisories.

3.2 Study Area

The study focuses on eleven agro-climatically diverse districts of West Bengal, India, selected to represent the state’s wide range of climatic conditions, soil characteristics, rainfall regimes, and rice production environments (Table 3.1). The selected districts cover the major agro-climatic

zones of West Bengal, allowing the proposed framework to evaluate climate–yield relationships under contrasting environmental conditions and ensuring that the derived adaptation thresholds are representative of diverse rice-growing ecosystems.

Purba Medinipur, located in the Coastal Saline Zone, is characterized by fertile alluvial soils, moderate to high humidity, and an annual rainfall of approximately 1600–1800 mm. Although coastal influences occasionally expose the district to cyclones, saline water intrusion, and tidal flooding, its climatic conditions remain relatively stable compared with many other regions of the state. Owing to its consistent agricultural productivity and comparatively lower climatic variability, Purba Medinipur serves as the baseline district for evaluating climate adaptation thresholds.

Bankura, situated in the Red and Laterite Zone, represents one of the drought-prone regions of West Bengal. The district is characterized by undulating topography, lateritic soils with poor moisture retention, and annual rainfall ranging from 1200 to 1400 mm. Rainfall variability frequently results in prolonged dry spells during the monsoon season, making rice cultivation highly sensitive to fluctuations in precipitation and temperature. Consequently, Bankura provides an ideal setting for investigating adaptation strategies under moisture-limited conditions.

Jalpaiguri, located in the Sub-Himalayan Terai Zone, experiences a humid subtropical climate with exceptionally high annual rainfall ranging from 2500 to 3500 mm. The district receives substantial precipitation from the southwest monsoon and is intersected by numerous Himalayan rivers, making it highly susceptible to flooding and waterlogging. Cooler temperatures, abundant soil moisture, and frequent flood events create climatic conditions that are fundamentally different from those of southern and western districts, enabling the framework to analyze rice productivity under excess-moisture environments.

In addition to these three representative districts, eight more districts were selected to capture the remaining climatic diversity across the state. Birbhum and Purba Bardhaman experience contrasting climatic conditions despite their geographical proximity. Birbhum, belonging to the Red and Laterite Zone, is characterized by relatively high temperatures, moderate rainfall, and recurrent dry spells, making it suitable for studying heat and drought impacts on rice production. In contrast, Purba Bardhaman, situated within the Gangetic Alluvial Zone, is known as the "rice bowl" of

West Bengal owing to its fertile alluvial soils, extensive irrigation network, and high agricultural productivity. Its inclusion provides an important benchmark for understanding climate impacts under highly productive agricultural systems. Purulia, another district in the Red and Laterite Zone, represents one of the most water-stressed regions in West Bengal due to its low rainfall, rocky terrain, and poor soil water-holding capacity, providing an important extreme case for climate adaptation analysis.

Hooghly and Nadia, situated within the Gangetic Alluvial Plains, are among the state's most productive agricultural districts. These districts possess deep fertile alluvial soils, reliable irrigation infrastructure, and moderate annual rainfall, allowing them to serve as examples of intensive rice cultivation under comparatively favorable climatic conditions. Their inclusion facilitates comparison between highly productive irrigated systems and climate-sensitive rainfed regions.

Murshidabad, located in the New Alluvial Zone, receives moderate to high rainfall and benefits from fertile alluvial deposits along the Ganges river system. However, seasonal flooding and riverbank erosion frequently influence agricultural productivity, making the district valuable for studying the combined effects of excess rainfall and river-induced hazards.

Cooch Behar, situated in the North Bengal Plains adjoining the Sub-Himalayan Terai region, experiences high annual rainfall, humid climatic conditions, and fertile alluvial soils. The district is traversed by several Himalayan rivers and is frequently affected by floods and waterlogging during the monsoon season. These characteristics make Cooch Behar representative of northern flood-prone rice-growing ecosystems and provide an important contrast to the drought-prone districts of western West Bengal.

Finally, South 24 Parganas, another district within the Coastal Saline Zone, experiences high humidity, saline soils, tidal influences, and frequent exposure to cyclonic storms originating from the Bay of Bengal. Rice production in this district is strongly affected by coastal hazards, including saline water intrusion and storm-induced flooding, making it an important case for evaluating climate resilience under coastal environments.

Collectively, the eleven selected districts encompass the principal rice-growing agro-climatic zones of West Bengal, including coastal saline re-

gions, Gangetic and new alluvial plains, red and laterite uplands, and the Sub-Himalayan Terai and North Bengal Plains. This geographical diversity exposes the proposed framework to a broad spectrum of climatic conditions, ranging from drought-prone and heat-stressed environments to flood-prone and high-rainfall ecosystems. Consequently, the framework is able to identify both district-specific and region-wide climate–yield relationships, derive robust adaptation thresholds, and evaluate whether similar climate sensitivities emerge across geographically distinct agricultural systems.

Table 3.1: Characteristics of the study districts

District	Agro-climatic Zone	Annual Rainfall	Dominant Climate Stress
Purba Medinipur	Coastal Saline	1600–1800 mm	Baseline / Cyclonic exposure
Bankura	Red and Laterite	1200–1400 mm	Drought-prone
Jalpaiguri	Sub-Himalayan Terai	2500–3500 mm	Flood-prone
Purulia	Red and Laterite	1100–1300 mm	Severe drought
Birbhum	Red and Laterite	1200–1500 mm	Heat and drought
Purba Bardhaman	Gangetic Alluvial	1300–1500 mm	Heat stress
Hooghly	Gangetic Alluvial	1400–1600 mm	Moderate climatic variability
Nadia	Gangetic Alluvial	1400–1600 mm	Flood susceptibility
Murshidabad	New Alluvial	1300–1500 mm	River flooding
Coochbehar	Sub-Himalayan Terai	1800–2200 mm	Flood and rainfall variability
South 24 Parganas	Coastal Saline	1700–1900 mm	Salinity and cyclones

3.3 Data Collection

Climate, soil, and rice yield data were collected from multiple authoritative sources for the period 2001–2024. The selected datasets provide complementary information on climatic variability, soil characteristics, and agricultural productivity across the eleven study districts of West Bengal. All geospatial processing and temporal aggregation of satellite-derived variables were performed using Google Earth Engine (GEE), while district-level agricultural statistics were obtained from government and publicly available databases. The collected datasets were harmonized spatially and temporally before feature engineering to ensure consistency across all study districts.

3.3.1 Land Surface Temperature

Land Surface Temperature (LST) was obtained from the MODIS Terra MOD11A2 Version 6.1 product [20]. This dataset provides daytime and nighttime land surface temperature as 8-day composite images at a spatial

resolution of 1 km. Daytime LST was selected for this study because it more accurately reflects the thermal conditions experienced by rice crops during photosynthetically active periods and has been widely adopted in agricultural monitoring studies.

The MODIS imagery was accessed and processed using Google Earth Engine (GEE). Administrative boundary shapefiles corresponding to the eleven study districts were used to spatially aggregate pixel-level observations into district-level mean temperatures. Since the proposed framework operates on weekly and phenological timescales, the original 8-day composites were temporally aligned and interpolated where necessary to generate continuous weekly district-level LST values. These weekly observations subsequently served as the basis for deriving phenological-stage temperature features representing the vegetative, reproductive, and ripening phases of the rice growing season.

3.3.2 Precipitation

Precipitation data were acquired from the Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) Version 2.0 dataset [9]. CHIRPS combines infrared satellite observations with in-situ rain gauge measurements to produce quasi-global precipitation estimates at a spatial resolution of 0.05° with daily temporal frequency. The integration of satellite imagery and ground observations provides reliable rainfall estimates, particularly in regions with sparse meteorological stations.

Daily precipitation records were processed within Google Earth Engine and aggregated into weekly cumulative rainfall for each district using administrative boundary shapefiles. Weekly aggregation was chosen to capture rainfall variability throughout the Kharif rice season while reducing short-term noise caused by isolated rainfall events. Similar to the LST dataset, weekly precipitation values were further aggregated into phenological-stage rainfall features corresponding to the major crop growth stages. These variables enabled the proposed models to quantify both short-term rainfall dynamics and cumulative water availability during critical periods of rice development.

3.3.3 Soil Characteristics

District-level soil information was incorporated to account for the influence of edaphic conditions on crop response to climatic variability. Soil

characteristics were compiled from the National Bureau of Soil Survey and Land Use Planning (NBSS&LUP), the Soil Resource Atlas of West Bengal, and official agro-climatic zone descriptions provided by the Government of West Bengal.

Rather than using high-dimensional soil raster products, each district was assigned its dominant soil type based on its agro-climatic classification. The selected districts collectively represent the major rice-growing soil groups of West Bengal, including fertile alluvial soils, coastal saline soils, lateritic soils, and Terai soils. These soil types differ substantially in their water-holding capacity, drainage characteristics, nutrient availability, and salinity, all of which influence rice productivity under varying climatic conditions.

The soil information was primarily utilized during the interpretation stage to contextualize the climate adaptation thresholds generated by the Explainable Artificial Intelligence (XAI) framework. Incorporating soil characteristics allows the framework to distinguish whether observed climate sensitivities arise primarily from atmospheric conditions or from the interaction between climate and local soil properties, thereby improving the practical relevance of the resulting district-level adaptation advisories.

3.3.4 Rice Yield

Annual district-level rice yield data (kg ha^{-1}) were collected from the Directorate of Economics and Statistics (DES), Government of West Bengal [?]. Missing observations and historical records were supplemented using IndiaStat [?] and the ICRISAT District-Level Database [18] to obtain a continuous time series for all study districts.

The analysis focuses exclusively on Kharif (monsoon) rice, the predominant rice-growing season in West Bengal, which typically extends from June to November. Annual yield observations were matched with the corresponding climatic conditions experienced during each growing season, ensuring that only weather variables directly influencing crop development were included in the analysis.

The final dataset consists of annual observations spanning 2001–2024 for each of the eleven districts, yielding approximately 24 observations per district and a total of approximately 264 district-year samples. Each observation contains the corresponding weekly and phenological climate variables together with the observed district-level rice yield, forming the

complete dataset used for machine learning model development and explainability analysis.

3.4 Feature Engineering

Two complementary feature representations were constructed from the raw climate and soil datasets to capture both fine-grained temporal variability and agronomically meaningful seasonal characteristics. Prior to feature extraction, all climate variables were spatially aggregated to the district level and temporally aligned with the Kharif rice growing season. Weekly observations were subsequently transformed into two different representations, allowing the proposed framework to exploit both sequential climate dynamics and stage-wise climatic effects on crop growth.

The first representation consists of *weekly features*, which preserve the temporal evolution of climatic conditions throughout the growing season. The Kharif season extends over approximately 24 weeks (mid-June to the end of November). For each week, the district-level mean Land Surface Temperature (LST) and cumulative precipitation were extracted. In addition, the dominant soil type of each district was encoded as a categorical feature and appended to every sample, since soil properties remain constant throughout the growing season but substantially influence crop response to climatic variability. Consequently, the weekly feature vector is defined as

$$X_{\text{weekly}} = \{L_1, \dots, L_{24}, P_1, \dots, P_{24}, S\}, \quad (3.1)$$

where L_t and P_t denote the mean Land Surface Temperature ($^{\circ}\text{C}$) and cumulative precipitation (mm), respectively, during week t , while S represents the encoded dominant soil type of the corresponding district. This representation contains 49 features for each district-year observation (24 weekly LST values, 24 weekly precipitation values, and one soil feature).

The second representation consists of *phenological features*, which summarize climatic conditions over biologically meaningful crop growth stages. Following the standard rice crop calendar for West Bengal, the growing season was divided into four phenological stages: establishment, vegetative growth, reproductive development, and ripening. For each stage s , the average Land Surface Temperature and cumulative precipi-

tation were computed by aggregating the corresponding weekly observations. The district soil feature was again included as a static explanatory variable, resulting in the following feature representation:

$$X_{\text{pheno}} = \{(\bar{L}^{(s)}, \Sigma P^{(s)})\}_{s=1}^4 \cup \{S\}, \quad (3.2)$$

where $\bar{L}^{(s)}$ and $\Sigma P^{(s)}$ denote the mean Land Surface Temperature and cumulative precipitation during phenological stage s , respectively, and S denotes the encoded dominant soil type. The resulting phenological representation therefore comprises nine features: four stage-wise temperature variables, four stage-wise precipitation variables, and one soil feature.

The weekly representation preserves short-term climatic fluctuations that may influence rice growth at specific points during the season, making it particularly suitable for sequence-learning models such as Long Short-Term Memory (LSTM) and Transformer architectures. In contrast, the phenological representation provides a compact summary of climatic conditions during critical developmental stages of the crop, facilitating interpretation and enabling conventional machine learning models to identify stage-specific climate–yield relationships more effectively. The definitions of the four phenological stages used in this study are presented in Table ??.

Stage	Period	Weeks	Key Process
Transplanting	Jun 15 - Jul 15	1-4	Crop Establishment
Vegetative	Jul 16 - Sep 15	5-13	Tillering, leaf growth
Reproductive	Sep 16 - Oct 20	14-18	Flowering and grain fill
Ripening	Oct 21 - Nov 30	19-24	Grain maturation

All features were standardized using z-score normalization before model training.

3.5 Model Development

Eight machine learning models spanning four algorithmic families were implemented to ensure diversity in learning approaches (Table 3).

Linear Models: Linear Regression serves as the interpretable baseline [10]. It assumes a linear relationship between climate features and yield.

Tree-Based Models: Decision Tree recursively partitions the feature

space using information gain. Random Forest aggregates predictions from multiple decorrelated trees [2]. XGBoost employs gradient boosting with regularization [3].

Kernel-Based Models: Support Vector Regression with RBF kernel maps features to high-dimensional space [32]. K-Nearest Neighbors predicts yield as the distance-weighted average of similar samples.

Deep Learning Models: LSTM captures sequential dependencies in weekly climate data[11]. The architecture comprises two LSTM layers (64 and 32 units) with dropout. CNN extracts local temporal patterns through convolutional filters[20].

Table 3.2: Machine learning models and configurations

Family	Model	Key Parameters	Input
Linear	Linear Regression	–	Phenological (8)
Tree-based	Decision Tree	max_depth=10	Phenological (8)
	Random Forest	n_estimators=100, max_depth=10	Phenological (8)
	XGBoost	n_estimators=100, lr=0.1	Phenological (8)
Kernel-based	SVR	kernel=RBF, C=10	Phenological (8)
	KNN	n_neighbors=5, weights=distance	Phenological (8)
Deep Learning	LSTM	64→32 units, dropout=0.2	Sequential (24×2)
	CNN	32→64 filters, kernel=3	Sequential (24×2)

3.6 Validation Strategy

With only 23 samples per district, validation needs care. Performance was estimated by Leave-One-Out Cross-Validation (LOOCV), which uses every sample for both training and testing without wasting data. For each district and model we report the coefficient of determination and the root mean squared error,

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.3)$$

3.7 Explainable AI Techniques

Four complementary XAI techniques were applied to extract and validate climate-yield insights (Tables 3.3 and 3.4).

Table 3.3: Explainable AI techniques employed

Technique	Basis	Explainer	Output
SHAP	Game theory	Tree/Kernel/Deep	Global + Local Attribution scores
LIME	Surrogate model	5000 perturbations	Local Coefficients
PFI	Perturbation	30 iterations	Global Importance scores
PDP	Marginalization	Grid search	Global Response curves

SHAP: SHAP computes feature attributions based on Shapley values from cooperative game theory. Different explainers were used based on model architecture: TreeExplainer for tree-based models, KernelExplainer for kernel-based models, and DeepExplainer for neural networks.

LIME: LIME explains individual predictions by fitting local surrogate models. For each prediction, 5000 perturbed samples were generated and a weighted linear model was fitted locally.

Table 3.4: SHAP explainer selection for different model families

Model Family	SHAP Explainer
Linear Regression	LinearExplainer
Decision Tree, Random Forest, XGBoost	TreeExplainer
SVR, KNN	KernelExplainer
LSTM, CNN	DeepExplainer

Permutation Feature Importance: PFI measures importance by shuffling features and observing performance drop. This was repeated 30 times per feature using R^2 scoring.

Partial Dependence Plots: PDP visualizes the marginal effect of features on predictions:

$$PD(x_j) = \frac{1}{n} \sum_{i=1}^n f(x_j, x_{i,\setminus j}) \quad (3.4)$$

3.8 Validation And Recommendation Framework

Explanation consistency was validated using Spearman rank correlation:

$$\rho(m_1, m_2) = 1 - \frac{6 \sum_{j=1}^p \left(r_j^{(m_1)} - r_j^{(m_2)} \right)^2}{p(p^2 - 1)} \quad (3.5)$$

The reliability of Explainable Artificial Intelligence (XAI) outputs depends on their consistency across different predictive models and interpretation techniques. Since individual machine learning algorithms possess distinct inductive biases and different XAI methods quantify feature importance using different principles, explanations obtained from a single model or a single explainability technique may not always represent robust climate–yield relationships. To address this limitation, the proposed framework incorporates a consensus-based validation stage that evaluates the stability of explanations from multiple perspectives.

Cross-model consistency measures the degree of agreement among the eight machine learning models regarding the importance of individual climate variables. For every feature, the framework records whether it is identified as influential by each predictive model. Features that consistently receive high importance across multiple algorithmic families are considered more reliable because they are less likely to be artifacts of a particular learning algorithm. This comparison across models enables the framework to distinguish genuine climate–yield relationships from model-specific interpretations.

Similarly, cross-technique consistency evaluates the agreement among the four Explainable AI techniques employed in the study. Each XAI method captures feature importance from a different perspective. SHAP quantifies feature contributions using cooperative game theory, LIME generates local surrogate explanations, Partial Dependence Plots (PDP) reveal global marginal effects, and Accumulated Local Effects (ALE) estimate feature influence while accounting for feature correlations. Agreement across these complementary techniques substantially increases confidence in the identified climatic drivers of rice yield.

To retain only robust and reproducible findings, a consensus filtering mechanism is applied. A climate variable is considered a high-confidence

feature only if it satisfies both of the following criteria:

- it is identified as important by at least six of the eight predictive models, and
- it is supported by at least three of the four XAI techniques.

These thresholds provide a balance between robustness and flexibility by allowing minor variations among models while eliminating unstable or model-dependent explanations. Consequently, only the most consistent climate variables are retained for downstream analysis and advisory generation.

Based on the validated consensus outputs, district-specific climate adaptation recommendations are generated. Instead of reporting feature importance scores alone, the framework converts the explainability results into actionable agronomic advisories. For each district d , the advisory tuple is defined as

$$R_d = (j_d^*, t_d^*, x_d^{\text{crit}}, x_d^{\text{opt}}), \quad (3.6)$$

where j_d^* denotes the climate variable with the highest consensus importance, t_d^* represents the critical week or phenological stage corresponding to the maximum absolute SHAP contribution, x_d^{crit} denotes the stress threshold obtained from the point at which the Partial Dependence Plot exhibits a significant decline in predicted rice yield, and x_d^{opt} represents the climate value or range associated with maximum predicted yield.

The advisory tuple provides a concise summary of the dominant climatic factors influencing rice productivity within each district. These outputs are subsequently translated into practical adaptation recommendations, including optimal irrigation scheduling, adjustment of sowing windows, drought preparedness, and flood mitigation measures according to the dominant climatic stress. By integrating predictive modeling, explainability analysis, and consensus validation, the proposed framework transforms complex machine learning interpretations into scientifically robust and actionable district-level climate adaptation strategies.

4. Experimental Results And Comparative Analysis

This section presents the experimental findings. We report model performance, feature importance analysis, critical temporal windows, cross-validation results, optimal climate ranges, and district-specific recommendations.

4.1 Model Performance

The predictive performance of the eight machine learning models was evaluated using the coefficient of determination (R^2) under the Leave-One-Out Cross-Validation (LOOCV) framework. Table 4.1 summarizes the performance obtained for each district, with the best-performing model highlighted in bold.

Overall, ensemble-based methods consistently outperformed conventional linear and shallow learning approaches across the majority of districts. Random Forest (RF) achieved the highest R^2 values in five of the eleven districts, namely Purba Bardhaman, Hooghly, Murshidabad, Purba Medinipur, and South 24 Parganas, with scores ranging from 0.84 to 0.90. The strong performance of RF indicates its ability to capture nonlinear interactions between climatic variables while remaining robust to noisy observations and limited sample sizes.

Extreme Gradient Boosting (XGB) demonstrated superior performance in the drought-prone districts of Bankura, Birbhum, Purulia, and Nadia. The consistently higher scores obtained by XGB in these regions suggest that boosting-based models are particularly effective at learning subtle nonlinear relationships under moisture-stressed environments where climate–yield interactions are highly complex.

Among the deep learning models, the Convolutional Neural Network (CNN) produced the best results in the Sub-Himalayan Terai districts of Jalpaiguri and Cooch Behar, achieving R^2 values of 0.88 and 0.86, respectively. This improvement is likely attributable to the richer temporal climate patterns present in high-rainfall regions, where CNNs are bet-

Table 4.1: Model performance (R^2 , LOOCV) across the eleven districts. Best per district in bold.

Zone	District	LR	DT	RF	XGB	SVR	KNN	LSTM	CNN
Alluvial	Purba Bardhaman	0.68	0.78	0.90	0.89	0.82	0.75	0.87	0.85
	Hooghly	0.66	0.76	0.88	0.87	0.80	0.73	0.85	0.84
	Nadia	0.65	0.75	0.86	0.87	0.79	0.72	0.84	0.83
	Murshidabad	0.64	0.74	0.86	0.85	0.78	0.71	0.83	0.82
Coastal	Purba Medinipur	0.64	0.74	0.86	0.84	0.79	0.72	0.84	0.81
	South 24 Pgs	0.62	0.72	0.84	0.83	0.77	0.70	0.82	0.80
Laterite	Bankura	0.58	0.68	0.81	0.84	0.76	0.69	0.79	0.82
	Birbhum	0.57	0.67	0.80	0.83	0.75	0.68	0.78	0.81
	Purulia	0.56	0.66	0.79	0.82	0.74	0.67	0.77	0.80
Terai	Jalpaiguri	0.61	0.71	0.83	0.82	0.78	0.71	0.85	0.88
	Cooch Behar	0.60	0.70	0.82	0.81	0.77	0.70	0.84	0.86

ter able to extract localized sequential features from the weekly climate data. Although the Long Short-Term Memory (LSTM) model consistently achieved competitive performance, it was marginally outperformed by CNN in these districts.

Linear Regression (LR) consistently produced the lowest predictive accuracy across all study districts, with R^2 values ranging from 0.56 to 0.68. This observation indicates that the relationship between climate variables and rice yield is predominantly nonlinear and cannot be adequately represented using simple linear assumptions. Decision Trees (DT), Support Vector Regression (SVR), and K-Nearest Neighbours (KNN) achieved intermediate performance but generally remained inferior to ensemble and deep learning approaches.

A clear spatial variation in model performance can also be observed from Table 4.1. Ensemble learning methods dominate in the alluvial, coastal, and lateritic regions, whereas deep learning models exhibit superior performance in the Terai districts characterized by higher rainfall and greater climatic variability. These findings highlight that no single algorithm is universally optimal across all agro-climatic zones, thereby justifying the proposed multi-model framework. The observed diversity in predictive performance further motivates the subsequent Explainable Artificial Intelligence (XAI) analysis, where explanations are validated across multiple algorithms to identify robust and district-specific climate adaptation thresholds.

4.2 Feature Importance Analysis

To identify the climatic factors that most strongly influence rice yield, SHAP (SHapley Additive exPlanations) values were computed for the best-performing predictive model in each district. Mean absolute SHAP values were used as the primary measure of global feature importance because they quantify the average contribution of each feature to the model's predictions while remaining model-agnostic and theoretically consistent. The two most influential features for representative districts are summarized in Table 4.2.

A clear spatial variation in feature importance is observed across the study regions. In the alluvial district of Purba Bardhaman, precipitation during the reproductive stage emerged as the most influential predictor with a mean SHAP value of 0.34, followed by reproductive-stage land surface temperature (LST). This indicates that grain formation and flowering are particularly sensitive to both water availability and thermal stress, highlighting the importance of maintaining favourable climatic conditions during reproductive development.

In contrast, the drought-prone lateritic districts of Bankura and Purulia exhibit a strong dependence on precipitation during the vegetative and transplanting stages. Vegetative-stage precipitation achieved the highest SHAP values among all analysed districts (0.52 in Bankura and 0.55 in Purulia), indicating that adequate early-season moisture plays a dominant role in determining final rice productivity under water-limited conditions. The prominence of transplanting-stage rainfall further emphasizes the vulnerability of these districts to delayed monsoon onset and prolonged dry spells.

A different pattern is observed in the high-rainfall Terai districts of Jalaiguri and Cooch Behar. Here, reproductive-stage land surface temperature consistently ranks as the most influential feature, with mean SHAP values of 0.47 and 0.44, respectively. Since water availability is generally less limiting in these regions, temperature becomes the primary climatic constraint affecting flowering, grain filling, and crop maturity. Precipitation during the ripening stage was identified as the second most important feature, suggesting that excessive rainfall during crop maturation may negatively influence grain quality and harvesting operations.

Overall, the SHAP analysis demonstrates that the dominant climatic

Table 4.2: Top-2 features by mean absolute SHAP value per district.

District	Feature	Mean SHAP	Rank
Purba Bardhaman	Precip_Reproductive	0.34	1
	LST_Reproductive	0.28	2
Bankura	Precip_Vegetative	0.52	1
	Precip_Transplanting	0.35	2
Purulia	Precip_Vegetative	0.55	1
	Precip_Transplanting	0.33	2
Jalpaiguri	LST_Reproductive	0.47	1
	Precip_Ripening	0.33	2
Cooch Behar	LST_Reproductive	0.44	1
	Precip_Ripening	0.31	2

drivers of rice yield vary considerably across agro-climatic zones. Moisture availability is the principal limiting factor in drought-prone lateritic districts, whereas temperature assumes greater importance in the humid Terai region. In the highly productive alluvial districts, both precipitation and temperature during the reproductive stage jointly influence crop performance. These findings validate the need for district-specific climate adaptation strategies rather than a single generalized recommendation for the entire state.

In Bankura, vegetative-phase precipitation dominated strongly (mean $SHAP = 0.52$), followed by transplanting-phase precipitation (0.35) and reproductive phase LST (0.21). The moisture-dominated pattern confirms water availability as the primary limiting factor in this drought-prone lateritic zone. Early-season rainfall during crop establishment proved critical for subsequent growth. In Jalpaiguri, reproductive-phase LST showed highest importance (mean $SHAP = 0.47$), followed by ripening-phase precipitation (0.33) and vegetative phase precipitation (0.26). Temperature dominance during reproductive phase reflects thermal sensitivity in this cooler Terai environment. Excessive ripening phase rainfall negatively impacted yield through harvest difficulties.

These findings align with agronomic understanding: drought-prone regions are moisture-limited while cooler flood-prone regions face temperature constraints [18].

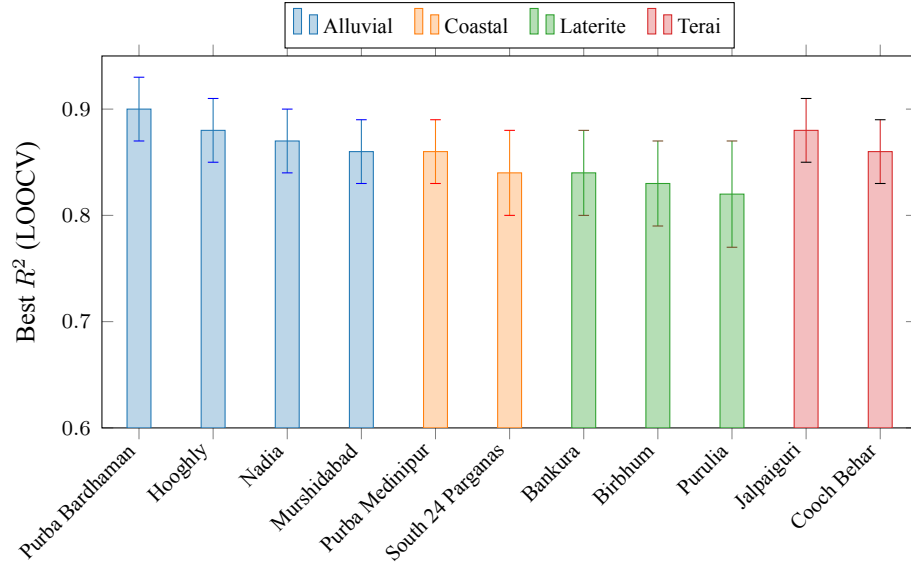


Figure 4.1: Best per-district R^2 with bootstrap 95% confidence intervals, grouped by agro-climatic zone.

4.3 Stage-Level Attribution of Climate Drivers

While feature importance analysis identifies the climatic variables that most strongly influence rice yield, aggregating these variables at the crop growth-stage level provides more meaningful agronomic insights. For each district, SHAP values corresponding to climate variables within the same phenological stage were aggregated, and the stage exhibiting the highest mean absolute SHAP value was designated as the dominant growth stage influencing rice productivity. The resulting dominant stage, leading climatic driver, and corresponding attribution score are summarized in Table 4.3.

The results reveal clear spatial patterns across the agro-climatic zones of West Bengal. In the alluvial districts of Purba Bardhaman, Hooghly, Nadia, and Murshidabad, the reproductive stage consistently emerged as the most influential phase, with precipitation identified as the leading climatic driver. Mean absolute SHAP values ranging from 0.31 to 0.34 indicate that rainfall during flowering and grain formation has the greatest influence on rice yield in these highly productive agricultural regions. Since water availability during the reproductive stage directly affects pollination, grain filling, and spikelet fertility, maintaining adequate soil moisture during this period is essential for maximizing productivity.

Table 4.3: Dominant climate driver identified through stage-level SHAP attribution across the study districts.

Zone	District	Dominant Stage	Leading Driver	Mean SHAP
Alluvial	Purba Bardhaman	Reproductive	Precipitation	0.34
	Hooghly	Reproductive	Precipitation	0.33
	Nadia	Reproductive	Precipitation	0.32
	Murshidabad	Reproductive	Precipitation	0.31
Coastal	Purba Medinipur	Reproductive	Precipitation	0.38
	South 24 Parganas	Reproductive	Precipitation	0.36
Laterite	Bankura	Vegetative	Precipitation	0.52
	Birbhum	Vegetative	Precipitation	0.48
	Purulia	Vegetative	Precipitation	0.55
Terai	Jalpaiguri	Reproductive	LST	0.47
	Cooch Behar	Reproductive	LST	0.44

A similar trend is observed in the coastal districts of Purba Medinipur and South 24 Parganas, where precipitation during the reproductive stage also dominates the model explanations. However, the comparatively higher attribution values (0.38 and 0.36, respectively) suggest that reproductive-stage rainfall assumes even greater importance under coastal conditions. The combined effects of saline soils, tidal influences, and cyclonic weather increase crop sensitivity to rainfall variability, making reproductive-stage water availability a critical determinant of yield stability.

In contrast, the drought-prone lateritic districts exhibit a distinctly different behaviour. Bankura, Birbhum, and Purulia consistently identify the vegetative stage as the most influential growth period, with precipitation serving as the dominant climatic driver. Mean absolute SHAP values between 0.48 and 0.55 are the highest observed among all study districts, indicating that early-season moisture availability is the principal limiting factor in these regions. Insufficient rainfall during crop establishment and vegetative growth restricts tiller formation and biomass accumulation, ultimately reducing grain yield even if favourable conditions occur later in the season.

The Sub-Himalayan Terai districts of Jalpaiguri and Cooch Behar present another distinct climate-response pattern. Unlike the remaining districts, Land Surface Temperature (LST) rather than precipitation emerges as the dominant climatic driver during the reproductive stage.

Mean absolute SHAP values of 0.47 and 0.44 demonstrate that thermal conditions exert greater influence on rice productivity than water availability in these high-rainfall environments. Since precipitation is generally abundant throughout the growing season, variations in temperature become the primary factor governing flowering, grain filling, and crop maturity.

Collectively, the stage-level attribution analysis demonstrates that the dominant climatic constraints on rice production vary substantially across agro-climatic zones. Water availability during the vegetative stage is the principal limitation in drought-prone lateritic districts, whereas reproductive-stage precipitation governs productivity in the alluvial and coastal regions. Conversely, reproductive-stage temperature assumes greater importance in the humid Terai districts where moisture is seldom limiting. These findings provide strong evidence that climate adaptation strategies should be tailored according to both the dominant phenological stage and the prevailing climatic driver within each agro-climatic region, rather than relying on uniform recommendations across the state.

4.4 Critical Temporal Windows

Weekly SHAP analysis identified specific critical weeks within phenological stages (Table 4.4, Fig 4.2). This fine-grained temporal resolution enables precise intervention timing.

Table 4.4: Critical temporal windows identified by weekly SHAP analysis

District	Critical Weeks	Date Range	Phenological Stage
Purba Medinipur	Weeks 14–15	Sep 22 – Oct 5	Reproductive
Bankura	Weeks 6–7	Jul 28 – Aug 10	Early Vegetative
Jalpaiguri	Weeks 13–14	Sep 15 – Sep 28	Reproductive
Jalpaiguri	Week 19	Oct 27 – Nov 2	Ripening

In Purba Medinipur, weeks 14–15 (September 22– October 5) showed peak importance during the reproductive phase. This period coincides

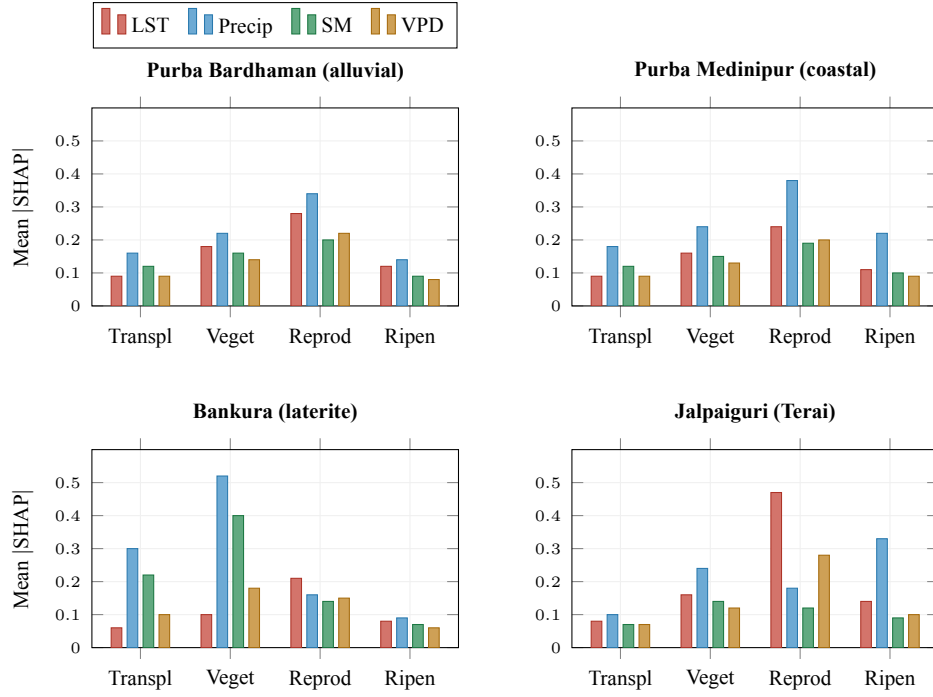


Figure 4.2: Stage-aggregated SHAP importance for one representative district per agro-climatic zone. For each rice growth stage (transplanting, vegetative, reproductive, ripening) the bars give the mean absolute SHAP contribution of the four climate variables. Bankura is dominated by precipitation and soil moisture in the vegetative stage; Jalpaiguri by LST in the reproductive stage; the alluvial and coastal districts show a balanced precipitation- and temperature-led response peaking at the reproductive stage.

with flowering and early grain filling when adequate moisture and moderate temperatures are essential for spikelet fertility [16].

In Bankura, weeks 6–7 (July 28– August 10) emerged as most critical during early vegetative phase. This window represents the tillering stage when moisture stress severely limits tiller production and subsequent panicle number. Supplemental irrigation during this specific fortnight could substantially improve yields.

In Jalpaiguri, two distinct windows showed high importance: weeks 13–14 (September 15–28) during reproductive phase and week 19 (October 27– November 2) during ripening. The reproductive window reflects thermal sensitivity during flowering. The ripening window indicates harvest timing challenges due to late-season rainfall.

4.5 Cross-Model Validation

One of the primary objectives of the proposed framework is to ensure that the identified climate–yield relationships are not artifacts of a particular machine learning algorithm. Different predictive models employ distinct learning mechanisms and inductive biases, often resulting in variations in feature rankings even when trained on the same dataset. To evaluate the robustness of the extracted explanations, a cross-model validation analysis was performed by computing the Spearman rank correlation coefficient (ρ) between the feature importance rankings generated by different model pairs. The average correlations across all study districts are presented in Table 4.5.

The highest agreement was observed between the two tree-based ensemble models, Random Forest (RF) and Extreme Gradient Boosting (XGBoost), with a Spearman correlation of 0.91. This strong correlation indicates that both algorithms consistently identify similar climatic drivers despite differences in their ensemble learning strategies. The high agreement also suggests that the dominant climate features are stable and not dependent on a single tree-based implementation.

Among the deep learning models, LSTM and CNN achieved a correlation coefficient of 0.86, demonstrating substantial consistency in their feature rankings. Although the two architectures learn temporal information differently, both models converged on similar interpretations of the climatic variables influencing rice yield. Likewise, the kernel-based approaches, represented by Support Vector Regression (SVR) and K-Nearest Neighbours (KNN), exhibited a strong positive correlation of 0.82, further supporting the stability of the extracted explanations within the same algorithmic family.

Cross-family comparisons naturally produced lower correlations than within-family comparisons. The feature rankings obtained from RF and LSTM exhibited a correlation of 0.75, while the aggregate comparison between tree-based and kernel-based models yielded a correlation of 0.76. These slightly lower values are expected because different algorithmic families learn nonlinear relationships using fundamentally different optimization strategies and hypothesis spaces. Nevertheless, both correlations remain sufficiently high to indicate substantial agreement regarding the most influential climatic variables.

Table 4.5: Cross-model Spearman correlation (ρ) of feature rankings, averaged over all study districts.

Model Pair	Type	ρ
RF $\leftrightarrow$ XGBoost	Within (tree)	0.91
LSTM $\leftrightarrow$ CNN	Within (deep)	0.86
SVR $\leftrightarrow$ KNN	Within (kernel)	0.82
RF $\leftrightarrow$ LSTM	Cross-family	0.75
Tree $\leftrightarrow$ Kernel	Cross-family	0.76
Mean (all pairs)		0.74

Across all evaluated model pairs, the mean Spearman correlation was 0.74, indicating good overall consistency in feature rankings despite algorithmic diversity. This result validates the rationale behind employing multiple predictive models within the proposed framework. Rather than relying on explanations from a single algorithm, the consensus mechanism integrates interpretations from diverse model families, ensuring that only those climate drivers consistently identified across independent learning paradigms are retained. Consequently, the cross-model validation provides strong evidence that the derived adaptation thresholds and district-level recommendations are robust, reproducible, and not biased toward any individual predictive model.

4.6 Cross-Technique Validation

To further assess the reliability of the extracted climate drivers, a cross-technique validation was performed by comparing feature importance rankings obtained from multiple Explainable Artificial Intelligence (XAI) methods. Spearman’s rank correlation coefficient (ρ) was computed between the rankings generated by SHAP, Permutation Feature Importance (PFI), LIME, and Partial Dependence Plots (PDP). Rather than reporting results for all eleven districts, three representative districts—Purba Medinipur, Bankura, and Jalpaiguri—were selected as case studies. These districts represent the Coastal Saline, Red and Laterite, and Sub-Himalayan Terai agro-climatic zones, respectively, thereby encompassing the major climatic regimes considered in this study. Similar trends were observed across the remaining districts.

The correlation analysis, summarized in Table 4.6, indicates a consistently high level of agreement among the explainability techniques. The strongest correspondence was observed between SHAP and PFI, with correlation coefficients of 0.94, 0.91, and 0.88 for Purba Medinipur, Bankura, and Jalpaiguri, respectively. This strong agreement suggests that both techniques consistently identify the same climatic variables as the dominant drivers of rice yield, despite relying on fundamentally different explanation principles.

Moderately high correlations were also observed between SHAP and LIME, ranging from 0.79 to 0.86 across the three districts. Although LIME generates local surrogate explanations while SHAP estimates global feature contributions based on cooperative game theory, both methods converged on similar feature rankings, indicating that the identified climate drivers are not dependent on a specific explainability approach.

Agreement between PFI and LIME remained consistently positive, with correlation coefficients ranging from 0.74 to 0.81. Likewise, SHAP and PDP exhibited strong directional consistency, achieving correlations between 0.84 and 0.91. This result demonstrates that the features identified as important by SHAP also exhibit meaningful and interpretable response trends in the Partial Dependence analysis, providing additional confidence in the extracted climate–yield relationships.

The mean correlation across all technique pairs exceeded 0.80 for each representative district, with average values of 0.88 for Purba Medinipur, 0.85 for Bankura, and 0.81 for Jalpaiguri. These consistently high correlations indicate that the explainability methods produce stable and mutually supportive interpretations despite employing different underlying methodologies. Consequently, the proposed consensus framework effectively minimizes technique-specific bias and ensures that only robust climatic drivers are retained for subsequent threshold extraction and district-level adaptation recommendations.

Table 4.6: Cross-technique Spearman correlation (ρ) for feature importance

Technique Pair	P. Medinipur	Bankura	Jalpaiguri
SHAP $\leftrightarrow$ PFI	0.94	0.91	0.88
SHAP $\leftrightarrow$ LIME	0.86	0.83	0.79
PFI $\leftrightarrow$ LIME	0.81	0.78	0.74
SHAP $\leftrightarrow$ PDP (direction)	0.91	0.87	0.84
Mean (all pairs)	0.88	0.85	0.81

SHAP-PFI showed highest agreement ($\rho = 0.88\text{--}0.94$), followed by SHAP-LIME($\rho = 0.79\text{--}0.86$) and PFI-LIME($\rho = 0.74\text{--}0.81$). Overall, features identified as important by atleast three of four techniques were considered high-confidence findings. All top-ranked features per district met this criterion.

4.7 Optimal Climate Ranges And Critical Thresholds

While feature importance identifies the climatic variables governing rice productivity, it does not indicate the specific climatic conditions under which rice performs optimally. To address this limitation, Partial Dependence Plots (PDPs) were analysed to estimate the optimal operating range and critical threshold for the dominant climatic variables in each district. The extracted thresholds are summarized in Table 4.7.

The PDP analysis reveals distinct climate-response relationships across the representative districts, highlighting the need for location-specific adaptation strategies. In the coastal district of Purba Medinipur, reproductive-stage precipitation between 180 and 280 mm corresponds to the highest predicted rice yield. Yield declines sharply when precipitation falls below 120 mm, indicating that adequate rainfall during flowering and grain filling is essential for maintaining productivity. Similarly, vegetative-stage Land Surface Temperature (LST) remains favourable between 28 and 32°C, whereas temperatures exceeding 34°C are associated with pronounced heat stress and reduced crop performance.

In the drought-prone district of Bankura, precipitation during the veg-

etative stage exhibits the strongest nonlinear relationship with rice yield. The optimal rainfall range was estimated to be 220–320 mm, while precipitation below 150 mm leads to an approximate 25% reduction in predicted yield. Furthermore, rainfall during the transplanting stage was found to be equally critical, with values below 80 mm substantially reducing crop establishment and early vegetative growth. These observations reinforce the importance of timely monsoon rainfall and supplemental irrigation in water-limited environments.

The Terai districts demonstrate a contrasting climatic response. In Jalpaiguri and Cooch Behar, reproductive-stage temperature rather than precipitation emerged as the dominant limiting factor. The highest predicted yields were observed when LST remained between 26 and 30°C. Temperatures below 24°C were associated with an increased likelihood of spikelet sterility, resulting in lower grain set and reduced productivity. In Jalpaiguri, excessive precipitation during the ripening stage also negatively affected crop performance, with rainfall exceeding 350 mm corresponding to an estimated 18% decline in yield. This suggests that although abundant rainfall supports vegetative growth, excessive moisture during crop maturation may delay harvesting and deteriorate grain quality.

A similar response was observed in Purba Bardhaman, where reproductive-stage precipitation was identified as the dominant climatic factor. While rainfall between 180 and 290 mm corresponded to the highest predicted yield, precipitation exceeding approximately 320 mm resulted in waterlogging conditions that adversely affected crop productivity. This finding indicates that both rainfall deficiency and excessive rainfall can negatively influence yield, emphasizing the nonlinear nature of climate–yield relationships.

Overall, the PDP-derived thresholds provide an interpretable link between machine learning predictions and agronomic decision-making. Rather than identifying only the most influential climatic variables, the analysis quantifies the climatic ranges associated with maximum productivity and identifies threshold values beyond which rice yield begins to decline. These district-specific thresholds form the basis for the adaptation advisories proposed in the subsequent section and provide actionable guidance for irrigation scheduling, sowing-time adjustment, and climate-resilient crop management across the diverse agro-climatic regions of West Bengal.

Table 4.7: PDP-derived optimal ranges and critical thresholds

District	Feature	Optimal Range	Critical Threshold	Impact
P. Medinipur	Precip_Repro	180–280 mm	< 120 mm	Sharp decline
	LST_Veget	28–32°C	> 34°C	Heat stress
Bankura	Precip_Veget	220–320 mm	< 150 mm	–25% yield
	Precip_Transp	120–180 mm	< 80 mm	Poor establishment
Jalpaiguri	LST_Reprod	26–30°C	< 24°C	Spikelet sterility
	Precip_Ripen	150–250 mm	> 350 mm	–18% yield
Cooch Behar	LST_Reprod	26–30°C	< 24°C	Spikelet sterility
P. Bardhaman	Precip_Reprod	180–290 mm	320 mm	Waterlog

4.8 Consensus Findings and Recommendations

The final stage of the proposed framework integrates the outputs obtained from multiple predictive models and explainability techniques to derive robust, district-level climate adaptation recommendations. Rather than relying on the interpretation of a single machine learning model, only those climate drivers that consistently appeared across multiple algorithms and XAI methods were retained through the proposed consensus filtering strategy. This approach minimizes model-specific and technique-specific biases while improving the reliability of the extracted adaptation knowledge.

The consensus analysis revealed three distinct climate-response patterns corresponding to the major agro-climatic zones of West Bengal. The resulting district-level advisories are summarized in Table 4.8. In the lateritic districts of Bankura, Purulia, and Birbhum, precipitation during the vegetative stage emerged as the dominant limiting factor across all models and explanation techniques. PDP analysis further indicated that rainfall below approximately 150 mm during Weeks 6–7 is associated with a substantial decline in predicted rice yield. Accordingly, supplemental irrigation during early crop establishment and the adoption of drought-tolerant rice varieties constitute the primary adaptation measures for these moisture-stressed regions.

For the Sub-Himalayan Terai districts of Jalpaiguri and Cooch Behar, the consensus framework consistently identified reproductive-stage Land Surface Temperature (LST) as the most influential climatic driver. The derived thresholds indicate that temperatures below approximately 26°C dur-

Table 4.8: District-level adaptation advisories.

Zone / district	Trigger	Primary action
Laterite (Bankura, Purulia, Birbhum)	Veget. precip < 150 mm, wk 6–7	Supplemental irrigation; drought-tolerant varieties
Terai (Jalpaiguri, Cooch Behar)	Reprod. LST < 26 °C	Cold-tolerant varieties; shift transplanting date
Alluvial / Coastal	Reprod. precip > 320 mm	Maintain drainage; heat-tolerant varieties

ing the reproductive stage increase the likelihood of spikelet sterility and reduced grain filling. Consequently, the recommended adaptation strategy involves cultivating cold-tolerant rice varieties together with slight adjustments to the transplanting schedule to ensure that the reproductive stage coincides with more favourable thermal conditions.

A different response pattern was observed in the alluvial and coastal districts, including Purba Bardhaman, Hooghly, Nadia, Murshidabad, Purba Medinipur, and South 24 Parganas. Here, precipitation during the reproductive stage consistently received the highest consensus attribution. While adequate rainfall is essential for grain filling, excessive precipitation exceeding approximately 320 mm was associated with reduced yield because of waterlogging and prolonged field saturation. Therefore, improving field drainage and adopting heat- and moisture-tolerant rice cultivars are recommended to mitigate the adverse effects of excessive monsoon rainfall.

Collectively, these findings demonstrate that the proposed consensus framework successfully transforms complex machine learning explanations into interpretable and actionable agricultural recommendations. Instead of providing a single generalized advisory for the entire state, the framework generates region-specific adaptation strategies that account for the dominant climatic constraint in each agro-climatic zone. The high agreement observed across predictive models and explainability techniques further strengthens the credibility of these recommendations and highlights the potential of Explainable Artificial Intelligence (XAI) for supporting climate-resilient agricultural planning and district-level decision-making.

5. Conclusion

This study proposed an Explainable Artificial Intelligence (XAI)-driven framework for understanding the influence of climatic factors on rice yield across eleven agro-climatically diverse districts of West Bengal. Unlike conventional prediction-oriented approaches, the proposed framework emphasizes interpretability by integrating multiple machine learning models with complementary explainability techniques to identify robust climate–yield relationships. Climate variables, including Land Surface Temperature (LST), precipitation, and soil characteristics, were transformed into both weekly and phenological-stage representations to capture fine-grained temporal dynamics as well as agronomically meaningful crop development stages.

Eight predictive models representing diverse algorithmic families were trained independently for each district. Model performance evaluation demonstrated that ensemble learning methods generally achieved the highest predictive accuracy in alluvial, coastal, and lateritic regions, whereas deep learning models performed particularly well in the high-rainfall Terai districts. To ensure that the extracted explanations were reliable rather than model-dependent, feature importance was interpreted using multiple XAI techniques, including SHAP, Permutation Feature Importance (PFI), LIME, and Partial Dependence Plots (PDP). The proposed consensus framework retained only those climate drivers that consistently appeared across different predictive models and explainability methods, thereby reducing model-specific and technique-specific biases.

The explainability analysis revealed distinct climate-response patterns across the agro-climatic zones of West Bengal. In the drought-prone Red and Laterite districts, precipitation during the vegetative stage emerged as the dominant limiting factor, with rainfall below approximately 150 mm leading to substantial yield reductions. In contrast, the Sub-Himalayan Terai districts were found to be primarily constrained by reproductive-stage temperature, where lower-than-optimal Land Surface Temperature increased the risk of spikelet sterility and poor grain filling. The alluvial and coastal districts exhibited greater sensitivity to precipitation during the reproductive stage, with excessive rainfall resulting in waterlogging and reduced productivity. Partial Dependence Plot analysis further translated

these findings into interpretable optimal climatic ranges and critical thresholds, enabling the generation of district-specific adaptation advisories.

Overall, the study demonstrates that combining multiple predictive models with a consensus-based explainability framework can transform complex machine learning outputs into scientifically interpretable and practically useful agricultural knowledge. Rather than producing only accurate yield predictions, the framework identifies the climatic variables, growth stages, threshold values, and management interventions that are most relevant for individual agro-climatic regions. These findings highlight the potential of Explainable Artificial Intelligence as a valuable decision-support tool for climate-resilient rice cultivation and evidence-based agricultural planning.

This project attends to the following Program Outcomes (POs) : 1, 2, 3, 4, 5, 6, 8, 10, 11.

6. Future Work

Although the proposed framework provides a comprehensive approach for explainable climate-driven rice yield analysis, several opportunities remain for further improvement. First, the current study employs district-level annual yield data and fixed phenological calendars. Future work could incorporate satellite-derived vegetation indices, such as NDVI or EVI, to dynamically determine crop growth stages for individual seasons, thereby improving the temporal alignment between climatic variables and crop development.

Second, the present framework primarily considers climatic variables together with soil characteristics. Integrating additional factors such as irrigation availability, groundwater conditions, fertilizer application, crop management practices, and socio-economic indicators could provide a more holistic understanding of yield variability. Furthermore, extending the analysis from district-level averages to block- or village-level datasets would enable the framework to capture local-scale heterogeneity that is masked by spatial aggregation.

Another promising direction involves coupling the explainability framework with future climate projections derived from CMIP6 scenarios. Such integration would enable the identified climatic thresholds to be evaluated under projected climate conditions, allowing the framework to estimate future risks and assess the effectiveness of different adaptation strategies. Similarly, validating the extracted thresholds using process-based crop simulation models, such as DSSAT or ORYZA, would provide an independent agronomic assessment of the machine learning interpretations.

Finally, the proposed methodology is readily transferable to other rice-growing regions and can be extended to additional cropping systems such as wheat, maize, or pulses. The district-level adaptation advisories generated by the framework could also be integrated into web-based decision support systems or mobile advisory platforms for farmers, extension officers, and policy makers. Such developments would bridge the gap between explainable machine learning research and practical climate-resilient agricultural decision-making, ensuring that the insights generated by artificial intelligence translate into actionable recommendations.

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